

ColliderAgent benchmark campaign, 16–26 September 2026

setup, protocol, every table and number for Tables S2–S5 and the Model C comparison

ColliderAgent benchmark harness, branch feat/skill-evolution-v2

2026-09-26

Part I — Tables S2–S5 of the Supplemental Material, filled, and the text to insert

The tables below follow the layout of sm.pdf (Tables S2–S5) so that they can be pasted into the source; the extended versions add the columns and rows measured beyond the paper’s placeholders. Every number is traceable to results-2026-09-26/runs.csv; Part II documents how it was obtained.

Table S2 — software environment

Caption as in the paper (“Software environment of the two container images used by Magnus in this work ...”). All versions were read inside the images on zhustation on 2026-09-16 with a probe blueprint. Two corrections to the placeholder table: uproot is not in the collider image (the analysis Python stack runs in the agent runtime), and Delphes is 3.5.1.

Image	Package	Version
collider	MadGraph5_aMC@NLO	3.7.0
	Pythia8	8.316
	Delphes	3.5.1
	MadAnalysis5	1.11.0
	LHAPDF	6.5.5
	numpy / scipy (in-image)	1.26.4 / 1.15.3
	awkward / matplotlib (in-image)	2.9.0 / 3.10.8
	pyhf	0.7.6
	Wolfram Engine	13.3.0
mma-het	FeynRules	2.3.49
agent runtime	uproot / awkward	5.7.4 / 2.9.0
	numpy / scipy	2.4.4 / 1.15.3
	Claude Code	2.1.273 (re-runs); original runs: TBD
	LLM	Claude Opus 4.6 (claude-opus-4-6)

Table S3 — resource usage per benchmark

Paper caption: “Resource usage per benchmark: wall-clock time from prompt to final figure, number of sub-agent invocations, Magnus jobs, files written by the agent (scripts, cards, records), and LLM tokens (input/output). Numbers refer to a successful run of each benchmark with Claude Opus 4.6.” Filled with the first successful attempt of each benchmark (2026-09-16, --effort xhigh); tokens = main session plus all sub-agents (uncached, cache-write and cache-read); [†] = no successful attempt yet, the numbers are those of the stopped (Scalar LQ, 6.4 h) or failed (mono- τ) attempt. The paper’s single row “U1 LQ at MuC [Fig. S2(e,f)]” is split into its two prompts; the Dark-SMEFT rows are the authors’ and unchanged.

Benchmark	Wall-clock [h]	Sub-agent calls	Magnus jobs	Files written	Tokens in [M]	Tokens out [k]
Scalar LQ, m_{ej} [Fig. 2(a)] [†]	6.41	2	42	28	214.0	220
ALP EFT, $\not{E}_T$ [Fig. 2(b)]	0.86	3	6	17	36.9	56
U(1)' scan [Fig. 2(c)]	1.05	3	17	25	5.2	16
U1 LQ mono- τ [Fig. 2(d)] [†]	4.00	2	29	19	181.4	219
Heavy N [Fig. S2(a,b)]	0.40	3	7	18	3.2	13
General Z' [Fig. S2(c)]	0.66	3	7	16	19.4	38
KK graviton [Fig. S2(d)]	1.35	2	16	14	4.2	51
U1 LQ at MuC, η [Fig. S2(e)]	0.55	0	14	2	1.3	5
U1 LQ at MuC, reach [Fig. S2(f)]	1.41	3	25	23	27.3	52
Dark-SMEFT, campaign 1 (O_{RV}, O_{RA})	≈ 24	TBD	>100	>200	TBD	TBD
Dark-SMEFT, campaign 2 (O_{LV}, O_{LA})	≈ 36	TBD	≈ 140	>200	TBD	TBD
Dark-SMEFT, campaign 3 (scalar/vector DM)	≈ 24	TBD	≈ 100	>100	TBD	TBD

Alternative filling (means over the three successful attempts of 2026-09-16 and 2026-09-24, same columns):

Benchmark	Wall-clock [h]	Sub-agent calls	Magnus jobs	Files written	Tokens in [M]	Tokens out [k]
Scalar LQ, m_{ej} [Fig. 2(a)] [†]	6.41	2.0	42.0	28.0	214.0	220
ALP EFT, $\not{E}_T$ [Fig. 2(b)]	1.01	3.0	6.7	16.7	15.1	31
U(1)' scan [Fig. 2(c)]	0.92	3.3	17.0	22.3	4.4	30
U1 LQ mono- τ [Fig. 2(d)] [†]	4.00	2.0	29.0	19.0	181.4	219
Heavy N [Fig. S2(a,b)]	0.60	3.0	9.0	14.3	3.9	17
General Z' [Fig. S2(c)]	0.73	3.0	7.3	14.3	8.3	25
KK graviton [Fig. S2(d)]	0.95	2.7	9.3	14.3	3.8	28
U1 LQ at MuC, η [Fig. S2(e)]	0.68	3.0	14.0	6.0	4.4	18
U1 LQ at MuC, reach [Fig. S2(f)]	1.22	3.0	19.0	21.7	13.3	39
Dark-SMEFT, campaign 1 (O_{RV}, O_{RA})	≈ 24	TBD	>100	>200	TBD	TBD
Dark-SMEFT, campaign 2 (O_{LV}, O_{LA})	≈ 36	TBD	≈ 140	>200	TBD	TBD
Dark-SMEFT, campaign 3 (scalar/vector DM)	≈ 24	TBD	≈ 100	>100	TBD	TBD

Model-latency share of wall-clock (transcript timestamps, 21 figure-benchmark attempts): 0.23–0.80, median 0.55.

Table S4 — reproduction success rate for different LLMs and agent run-times

Paper caption: “Reproduction success rate (successful runs / attempted runs) per benchmark for different LLMs and agent runtimes.” Model B = GPT-5.3-Codex (released 2026-02-05, the same day as Opus 4.6) through the Codex CLI in skills-only mode (branch codex-com); model C = Gemini 3.1 Pro preview (2026-02-19) through the single-agent ADK reference implementation (python-agent). Paper layout (the four main-text benchmarks):

Benchmark	Opus 4.6	model B: GPT-5.3-Codex (Codex)	model C: Gemini 3.1 Pro (ADK)
Scalar LQ, m_{ej}	0/1 (1 TBD)	0/1	0/1
ALP EFT, $\not{E}_T$	3/3	1/1	2/3
U(1)' scan	3/3	1/1	1/3
U1 LQ mono- τ	0/1	1/1	0/1

Extended version (all benchmarks; every LLM/runtime pair measured in the campaign; $\times 3$ where three attempts were made):

Benchmark	Opus 4.6 Claude Code	Opus 5 Claude Code	GPT-5.3-Codex Codex	GPT-5.5 Codex	Gemini 3.1 Pro ADK	Gemini 3.1 Pro Gemini CLI	GPT-5.5 ADK
Scalar LQ, m_{ej} [Fig. 2(a)]	0/1 (1 TBD)	–	0/1	1/1	0/1	–	0/1
ALP EFT, $\not{E}_T$ [Fig. 2(b)]	3/3	1/1	1/1	1/1	2/3	1/1	1/1
U(1)' scan [Fig. 2(c)]	3/3	1/1	1/1	1/1	1/3	1/1	1/1
U1 LQ mono- τ [Fig. 2(d)]	0/1	–	1/1	1/1	0/1	–	0/1
Heavy N [Fig. S2(a,b)]	3/3	1/1	1/1	–	1/3	1/1	1/1
General Z' [Fig. S2(c)]	3/3	1/1	1/1	–	1/3	1/1	1/1
KK graviton [Fig. S2(d)]	3/3	1/1	1/1	–	2/3	1/1	1/1
U1 LQ at MuC, η [Fig. S2(e)]	2/3	1/1	0/1	–	0/3	1/1	1/1
U1 LQ at MuC, reach [Fig. S2(f)]	3/3	1/1	0/1	–	1/3	1/1	1/1

Table S5 — reproduction robustness

Paper caption: “Reproduction robustness: number of successful runs out of three sandboxed attempts per benchmark, and the failure mode of the unsuccessful runs, classified as (i) model, (ii)

generation, (iii) analysis or (iv) infrastructure.” Opus 4.6, Claude Code; the two Delphes-level benchmarks have one attempt each (see Part II, open items):

Benchmark	Successful / attempted	Failure mode(s)
Scalar LQ, m_{ej}	0/1	stopped by the operator (TBD)
ALP EFT, $\not{E}_T$	3/3	-
U(1)' scan	3/3	-
U1 LQ mono- τ	0/1	(iv) infrastructure
Heavy N	3/3	-
General Z'	3/3	-
KK graviton	3/3	-
U1 LQ at MuC, η	2/3	(i) model
U1 LQ at MuC, reach	3/3	-

Extended version (the two runtimes with three attempts per benchmark; the ADK loop’s attempts include its two Delphes-level runs):

Benchmark	Opus 4.6 / Claude Code		Gemini 3.1 Pro / ADK	
	succ./att.	failure mode(s)	succ./att.	failure mode(s)
Scalar LQ, m_{ej}	0/1	stopped by the operator (TBD)	0/1	(ii) generation
ALP EFT, $\not{E}_T$	3/3	-	2/3	(i) model
U(1)' scan	3/3	-	1/3	(ii) generation $\times 2$
U1 LQ mono- τ	0/1	(iv) infrastructure	0/1	(ii) generation
Heavy N	3/3	-	1/3	(i) model $\times 2$
General Z'	3/3	-	1/3	(iii) analysis, (i) model
KK graviton	3/3	-	2/3	(ii) generation
U1 LQ at MuC, η	2/3	(i) model	0/3	(ii) generation $\times 2$, (i) model
U1 LQ at MuC, reach	3/3	-	1/3	(iii) analysis, (ii) generation

Text to insert (reply letter and Supplemental Material)

Where	Placeholder	Proposed text
reply p. 1, check-list (1)	fill in Tables S3, S4, S5	Done: the tables above; the numbers of this letter follow.
reply p.4, point (c)	"[choose one of the two sentences]"	"We have therefore run all eight benchmarks with GPT-5.3-Codex through the Codex CLI (skills-only mode) and with Gemini 3.1 Pro through the single-agent ADK reference implementation, three attempts each for Opus 4.6 and for the ADK runtime, and report the success rates in Table S4 of the SM. The same Gemini model succeeds in 7/7 figure benchmarks inside a coding-agent runtime (Gemini CLI, skills-only) but in 8/21 attempts inside the single-agent loop, and no single-agent run completed a Delphes-level benchmark, so the reliability with which the skills are followed depends on the runtime at least as much as on the model."
reply p.4, "Tokens and latency"	"the LLM token consumption [fill in]"	"the LLM token consumption (1-37 M input and 5-56 k output tokens per figure benchmark, 180-215 M input tokens for the two Delphes-level benchmarks, main session plus sub-agents)". Next sentence: "The wall-clock time of a benchmark (0.4-1.4 h for the figure benchmarks, 4-6 h for the Delphes-level ones) is shared between LLM inference and cluster jobs for the former (model share 0.2-0.8, median 0.55) and dominated by event generation and detector simulation for the latter."
reply p.4, "Robustness"	sentence on three runs and Table S5	keep; Table S5 is filled (20/21 for the seven figure benchmarks; the two Delphes-level benchmarks have one attempt each, see the open items).
reply p.6	"Table S3 quantifies ...[fill in]"	"Table S3 quantifies, per benchmark, the number of sub-agent invocations (2-3), Magnus jobs (6-42), files written (2-28) and wall-clock time (0.4-6.4 h)".
sm p.5, Sec. S-I.E	"LLM inference accounts for a minor fraction [fill in]"	"For the figure benchmarks of Table S3 the wall-clock time is shared about equally between LLM inference and cluster jobs (model share of wall-clock 0.2-0.8, median 0.55 over 21 runs); cluster time dominates only for the Delphes-level benchmarks and the Dark-SMEFT campaigns."
sm p.5, Sec. S-I.E	Table S4 placeholder sentence	"Table S4 reports the reproduction success rate of the benchmarks for Claude Opus 4.6 in Claude Code (full multi-agent mode), GPT-5.3-Codex in the Codex CLI (skills-only mode) and Gemini 3.1 Pro in the ADK reference implementation (single agent); the extended version in the results package adds Opus 5, GPT-5.5 and Gemini CLI."
sm p.6, Table S3 caption	"[fill in from usage logs]"	"Numbers refer to the first successful run of each benchmark with Claude Opus 4.6 (effort xhigh); tokens count the main session and all sub-agents."
sm p.6, Table S5 and text	"[fill in the per-benchmark counts and failure modes]"	Table S5 above; sentence: "Across the seven figure benchmarks, 20 of 21 sandboxed attempts produced the requested figure; the one failure was a model-construction failure (the leptoquark coupling did not enter the generated sample). The two Delphes-level benchmarks failed once each on infrastructure (job storage limit; an early stop while waiting for downloads) and are being re-run."
paper p.7, Table I caption	"tokens consumed and wall-clock time [fill in]"	cite Table S3 of the SM instead of repeating numbers.
sm p.4, Table S2	Delphes "3.5.x (TBD)", MadAnalysis5 TBD, analysis stack TBD, Claude Code TBD	rows above; the Claude Code version of the original April-May runs must come from the co-author who ran them.

Part II — the campaign record

1 Scope and timeline

The paper’s supplementary tables S2 (software versions), S3 (resource usage), S4 (models and harnesses) and S5 (robustness, three attempts) were empty. Between 2026-09-16 and 2026-09-26 the nine paper-reproduction prompts of the repository were run under five harness/model combinations, judged against the reference figures, and their resource usage recorded. This document is the single record of that campaign; the per-run data are in docs/paper/results-2026-09-24/ and docs/paper/results-2026-09-26/ (runs.csv, tables.md, figures/).

Date (UTC)	What ran
09-16	Magnus site zhustation verified; skills rewritten (v2, context-lean, sidecars, run-lessons); Table S2 read inside the images with a probe blueprint; ALP EFT run with Opus 5, Opus 4.8, Sonnet 5 (1 each); Opus 4.6 xhigh attempt 1 of all nine prompts (22:00 UTC).
09-17	Attempt 1 judged: 7/9, Scalar LQ stopped at 6.4 h (27 GB Delphes ROOT vs 10 GB job space), mono- τ early stop; Opus 5 cross-check of U(1)'; handover written.
09-18	Memory/evolution audit; parameter-error notes into skills; dependency fixes.
09-24	Opus 4.6 attempts 2-3 of the seven figure benchmarks (14 valid runs; 4 runs lost to copied OAuth tokens and relaunched); Codex + GPT-5.5 (ChatGPT login) on ALP, U(1)', mono- τ , Scalar LQ; results package v1.
09-26	Model C: python-agent (Google ADK single loop) + Gemini 3.1 Pro, 7 figure benchmarks $\times 3$ and both Delphes benchmarks; controls Gemini CLI + Gemini 3.1 Pro (7) and ADK + GPT-5.5 (7 + 2); Codex + GPT-5.3-Codex via API key (7 + 2); Opus 5 on the remaining 5 figure benchmarks; job-provenance check introduced; results package v2.

2 Benchmarks

Nine prompts from paper-reproduction/<arxiv>/prompt_figure_<n>.md; seven are “figure” benchmarks (model $\rightarrow$ UFO $\rightarrow$ MadGraph parton level or scan $\rightarrow$ analysis and plot) and two are “Delphes” benchmarks (100k-event Pythia8+Delphes samples, LHE rewriting, job-size limits).

Paper	Figure	Task	Class
1308.2209	Fig. 3	Heavy Majorana neutrino: $\sigma(pp \rightarrow \mu^\pm N)$ vs m_N at 7/8/14 TeV	figure
1605.02910	Fig. 1	U(1)' Z': LHC Run-1 dilepton 2σ exclusion contours in $(\tilde{g}, g_1')$ for $M_{Z'} = 2, 2.5, 3$ TeV	figure
1701.05379	Fig. 8	ALP EFT: normalised E_T^{miss} distribution of $pp \rightarrow aW^\pm\gamma$	figure
2103.02708	Fig. 4	General Z': $\sigma(pp \rightarrow Z' \rightarrow \mu\mu)$ vs mass for SSM and ψ benchmarks	figure
9909255	Fig. 2	KK graviton: $\sigma(e^+e^- \rightarrow \mu^+\mu^-)$ vs $\sqrt{s}$	figure
2104.05720	Fig. 11	U1 leptoquark at a muon collider: normalised η distributions, two couplings	figure
2104.05720	Fig. 12	U1 leptoquark at a muon collider: exclusion/discovery contours in (m_{LQ}, β_L^{32})	figure
2005.06475	Fig. 2	Scalar leptoquark: m_{ej} distribution with LUXlep lepton PDF, Pythia8+Delphes, selection	Delphes
1811.07920	Fig. 3	U1 leptoquark mono- τ : 2σ exclusion in $(\sqrt{ g_c g_b }, M_{U_1})$, 18 Delphes samples	Delphes

3 Infrastructure (Table S2)

All compute runs on the Magnus platform (magnus CLI 0.8.1-0.8.2, site zhustation); eight blueprints (validate-feynrules, generate-ufo, generate-calcchep, madgraph-compile, madgraph-launch, madanalysis-process, micromegas-compile, micromegas-calc). Each job has 10 GB of packing space. Versions read inside the images on 2026-09-16:

Image / runtime	Package	Version	Evidence
collider	MadGraph5_aMC@NLO	3.7.0	/opt/MG5_aMC_v3_7_0
collider	Pythia8	8.316	pythia8-config --version
collider	Delphes	3.5.1	installer tarball Delphes-3.5.1
collider	MadAnalysis5	1.11.0 (2025-04-23)	/opt/madanalysis5/version.txt
collider	LHAPDF	6.5.5	lhpdf-config --version
collider	HepMC	2.06.09	event-file header
collider	ROOT	6.28/10	root-config --version
collider	Python	3.10.12; numpy 1.26.4, scipy 1.15.3, awkward 2.9.0, matplotlib 3.10.8, pyhf 0.7.6	in-image import
mma-het	Wolfram Engine	13.3.0	.VersionID
mma-het	FeynRules	2.3.49 (29 Sep 2021)	FR\$VersionNumber
micromegas	micrOmegas / CalcHEP	6.3.0 / 3.9.2	image rise-agi/micromegas
agent host	Python analysis stack	3.13.12; uproot 5.7.4, awkward 2.9.0, numpy 2.4.4, scipy 1.15.3, matplotlib 3.10.9, pyhf 0.7.6	host import
agent host	Claude Code	2.1.273 (09-16/17), 2.1.282 (09-24), 2.1.28x (09-26)	claude --version
agent host	Codex CLI	0.155.1	codex --version
agent host	Gemini CLI	0.61.0	gemini --version
agent host	Google ADK / LiteLLM	2.10.0 / 1.x (python-agent venv)	uv sync

4 Harnesses, models and their structure

Column	Model (release)	Harness	Structure and budget
A	Claude Opus 4.6 (2026-02-05), --effort xhigh	Claude Code, v2 skills	orchestrator + 4 stage sub-agents (memory: user), 13 skills, step sidecars, run-lessons memory; no call cap
A'	Claude Opus 5 (default effort)	same	same
B	GPT-5.3-Codex (2026-02-05)	Codex CLI 0.155.1, API key, branch codex.com	skills-only (.agents/skills), 4 custom agents (.codex/agents), AGENTS.md; no call cap
B'	GPT-5.5 (2026-04-23)	Codex CLI, ChatGPT login	same
C	Gemini 3.1 Pro preview (2026-02-19) via OpenAI-compatible relay	python-agent: Google ADK 2.10 single LLMAgent + LiteLLM	13 flat tools (read/write/edit, validate, UFO, compile, launch, YAML workflow, MadAnalysis, run_python, run_shell); no sub-agents, skills, memory or context compaction; max_llm_calls 150 (300 for Delphes benchmarks), 6 h wall
C ₁	Gemini 3.1 Pro preview (native API through the relay)	Gemini CLI 0.61, skills-only	same .agents/skills as B, GEMINI.md; no custom sub-agents; --yolo
C ₂	GPT-5.5 (OpenAI API)	ADK, identical code to C	as C

5 Protocol

- One clean sandbox per attempt (scripts/bench/run_benchmark*.sh); the prompt is the benchmark file plus, from 2026-09-26, scripts/bench/benchmark_rules.md (submit your own Magnus jobs; never read or download jobs this session did not submit). Cold memory for every run.
- Success (agreed 2026-09-16): the run completes end to end and the figure agrees with the requested part of the reference qualitatively; quantitative deviations go to a footnote (*). Failure modes follow the paper: model (m), generation (g), analysis (a), infrastructure (i). An assistive judge (scripts/bench/judge.sh, Opus 5, target-aware, lenient prompt) proposes the verdict; every verdict was confirmed by eye, and overridden where the judge missed a wrong pipeline (parton-level substitutes for Delphes, inverted benchmark curves).
- Metrics: wall-clock from the harness; tokens = main session plus all sub-agents (Claude: transcripts; Codex: main and sub-agent rollouts; Gemini CLI: stream-json stats; ADK: per-event usage); Magnus jobs = magnus run submissions or ADK tool calls to the five Magnus-backed tools; USD only for Claude runs (CLI total_cost_usd); ADK additionally records LLM calls, tool errors, error-refine cycles (a failed tool call followed by another call of the same tool), the largest prompt and the exit reason.
- Provenance (from 2026-09-26): scripts/bench/job_provenance.py scans each sandbox (and Codex sub-agent rollouts) for referenced vs submitted job ids; a run using foreign jobs is invalid.

6 Results

6.1 Table S4: successful/attempted per benchmark and column

Opus 4.8 and Sonnet 5 (one ALP EFT run each, both successful) are listed in §6.4.

Benchmark	Opus 4.6 Claude Code	Opus 5 Claude Code	GPT-5.3-Codex Codex	GPT-5.5 Codex	Gemini 3.1 Pro ADK	Gemini 3.1 Pro Gemini CLI	GPT-5.5 ADK
Heavy N	3/3	1/1	1/1	-	1/3	1/1	1/1
U(1)' scan	3/3	1/1	1/1	1/1	1/3	1/1	1/1
ALP EFT	3/3	1/1	1/1	1/1	2/3	1/1	1/1
General Z'	3/3	1/1	1/1	-	1/3	1/1	1/1
KK graviton	3/3	1/1	1/1	-	2/3	1/1	1/1
U1 LQ at MuC, η	2/3	1/1	0/1	-	0/3	1/1	1/1
U1 LQ at MuC, reach	3/3	1/1	0/1	-	1/3	1/1	1/1
Scalar LQ m_{ej} (Delphes)	0/1 (1 TBD)	-	0/1	1/1	0/1	-	0/1
U1 LQ mono- τ (Delphes)	0/1	-	1/1	1/1	0/1	-	0/1
figure benchmarks	20/21	7/7	5/7	2/2	8/21	7/7	7/7

6.2 Table S5: per-attempt results and failure modes

Columns with three attempts (ok = success, * = with footnote; m/g/a/i = failure mode; TBD = stopped by the operator):

Benchmark	Opus 4.6 / Claude Code (a1 / a2 / a3)	Gemini 3.1 Pro / ADK (a1 / a2 / a3)
Heavy N	ok / ok / ok*	m / m / ok*
U(1)' scan	ok* / ok* / ok*	g / g / ok*
ALP EFT	ok / ok* / ok*	ok* / m / ok*
General Z'	ok / ok* / ok*	a / m / ok*
KK graviton	ok* / ok* / ok*	ok* / ok* / g
U1 LQ at MuC, η	ok* / m / ok*	g / m / g
U1 LQ at MuC, reach	ok* / ok* / ok*	a / g / ok*
Scalar LQ m_{ej} (Delphes)	TBD	g
U1 LQ mono- τ (Delphes)	i	g

Failure-mode counts over every run of the campaign (figure and Delphes benchmarks):

Column	runs	ok	TBD	model	generation	analysis	infrastructure
Opus 4.6 / Claude Code	23	20	1	1	0	0	1
Opus 5 / Claude Code	7	7	0	0	0	0	0
Opus 4.8 / Claude Code	1	1	0	0	0	0	0
Sonnet 5 / Claude Code	1	1	0	0	0	0	0
GPT-5.3-Codex / Codex	9	6	0	1	1	1	0
GPT-5.5 / Codex	4	4	0	0	0	0	0
Gemini 3.1 Pro / ADK	23	8	0	5	8	2	0
Gemini 3.1 Pro / Gemini CLI	7	7	0	0	0	0	0
GPT-5.5 / ADK	9	7	0	0	2	0	0

6.3 Table S3: resource usage

Every Opus 4.6 attempt (attempt 1 of 2026-09-16 from the documented numbers, attempts 2-3 of 2026-09-24 from the sandboxes; model share = model latency / wall-clock from transcript timestamps):

Benchmark	attempt	wall [h]	sub-agents	jobs	files	tok in [M]	tok out [k]	USD	model share	result
Heavy N	1 (09-16)	0.40	3	7	18	3.2	13	4.2	0.72	ok
	2 (09-24)	0.37	2	7	13	2.7	16	3.6	0.54	ok
	3 (09-24)	1.04	4	13	12	5.7	22	6.7	0.33	ok*
U(1)' scan	1 (09-16)	1.05	3	17	25	5.2	16	9.2	0.69	ok*
	2 (09-24)	1.07	4	17	21	4.5	55	7.5	0.54	ok*
	3 (09-24)	0.65	3	17	21	3.6	18	6.0	0.69	ok*
ALP EFT	1 (09-16)	0.86	3	6	17	36.9	56	22.2	0.78	ok
	2 (09-24)	1.30	3	8	20	5.1	19	10.5	0.52	ok*
	3 (09-24)	0.87	3	6	13	3.2	17	5.7	0.46	ok*
General Z'	1 (09-16)	0.66	3	7	16	19.4	38	13.3	0.80	ok
	2 (09-24)	0.59	3	6	14	2.5	19	3.9	0.44	ok*
	3 (09-24)	0.94	3	9	13	2.9	19	4.4	0.23	ok*
KK graviton	1 (09-16)	1.35	2	16	14	4.2	51	6.4	0.33	ok*
	2 (09-24)	0.76	3	7	14	3.6	15	6.1	0.63	ok*
	3 (09-24)	0.75	3	5	15	3.6	18	6.6	0.66	ok*
U1 LQ at MuC, η	1 (09-16)	0.55	0	14	2	1.3	5	5.9	0.47	ok*
	2 (09-24)	0.74	0	11	8	21.1	79	14.5	0.55	m
	3 (09-24)	0.81	6	14	10	7.5	32	9.7	0.78	ok*
U1 LQ at MuC, reach	1 (09-16)	1.41	3	25	23	27.3	52	20.6	0.64	ok*
	2 (09-24)	1.38	3	19	25	8.2	28	10.0	0.48	ok*
	3 (09-24)	0.87	3	13	17	4.2	38	7.4	0.67	ok*
Scalar LQ m_{ej} (Delphes)	1 (09-16)	6.41	2	42	28	214.0	220	-	0.65	TBD
U1 LQ mono- τ (Delphes)	1 (09-16)	4.00	2	29	19	181.4	219	106.0	0.68	i

Means over successful figure-benchmark runs per column (files = distinct paths written through the harness's file tools, not comparable for Codex which writes through shell heredocs; USD only where the CLI reports it):

Column	ok runs	wall [h]	sub-agents	Magnus jobs	files	tokens in [M]	tokens out [k]	USD
Opus 4.6 / Claude Code	20	0.88	3.0	11.7	16.1	7.8	27	8.5
Opus 5 / Claude Code	7	1.56	3.4	10.6	10.6	9.3	52	13.9
Opus 4.8 / Claude Code	1	1.05	3.0	9.0	16.0	11.0	43	12.6
Sonnet 5 / Claude Code	1	1.38	3.0	14.0	24.0	18.2	34	7.5
GPT-5.3-Codex / Codex	5	0.84	3.0	15.6	0.2	5.9	41	-
GPT-5.5 / Codex	2	1.01	3.0	15.0	12.5	12.2	52	-
Gemini 3.1 Pro / ADK	8	1.17	0.0	23.6	0.5	4.5	29	-
Gemini 3.1 Pro / Gemini CLI	7	0.63	0.0	11.3	6.0	3.7	15	-
GPT-5.5 / ADK	7	0.71	0.0	13.4	2.1	1.9	19	-

Model-latency share of wall-clock, Opus 4.6, 21 figure-benchmark attempts: 0.23–0.80, median 0.55; above 0.6 in 7 of 9 attempt-1 runs. Cost of the campaign in Claude CLI dollars: Opus 4.6 attempt 1 $\approx$ 190 (Scalar LQ excluded), attempts 2–3 102.6, Opus 5 $\approx$ 65; Codex, Gemini and ADK ran on API keys or a relay without per-run billing.

6.4 Single cross-model runs of 2026-09-16 (default effort)

Benchmark	model	wall [h]	sub-agents	jobs	files	tok in [M]	USD	result
U(1)' scan	Opus 5 / Claude Code	0.88	4	7	16	9.1	14.32	ok*
ALP EFT	Opus 4.8 / Claude Code	1.05	3	9	16	11.0	12.55	ok
ALP EFT	Opus 5 / Claude Code	0.87	3	5	17	5.7	9.06	ok
ALP EFT	Sonnet 5 / Claude Code	1.38	3	14	24	18.2	7.54	ok*

6.5 Long-horizon indicators

Means per run; LLM calls, tool errors, error-refine cycles and the largest context are only measurable for the ADK harness (its event log records every model call and every tool's success flag). Figure benchmarks:

Column	n	ok	wall [h]	LLM calls	tool calls	tool err.	refine cyc.	jobs	max ctx [k]	tok in [M]	exit
Opus 5 / Claude Code	5	5	1.83	-	-	-	-	12	-	10.1	-
Gemini 3.1 Pro / ADK	21	8	1.41	117	117	39	39	26	120	8.3	completed (11), max_llm_calls (10)
Gemini 3.1 Pro / Gemini CLI	7	7	0.63	-	72	-	-	11	-	3.7	completed (7)
GPT-5.3-Codex / Codex	7	5	0.81	-	-	-	-	15	-	5.3	-
GPT-5.5 / ADK	7	7	0.71	35	39	4	4	13	71	1.9	completed (7)

Delphes benchmarks:

Column	n	ok	wall [h]	LLM calls	tool calls	tool err.	refine cyc.	jobs	max ctx [k]	tok in [M]	exit
Gemini 3.1 Pro / ADK	2	0	2.04	300	300	52	50	28	169	28.4	max_llm_calls (2)
GPT-5.3-Codex / Codex	2	1	1.72	-	-	-	-	13	-	9.5	-
GPT-5.5 / ADK	2	0	1.22	54	57	13	12	18	91	3.7	completed (2)

Every ADK run (transcribed from each sandbox’s `metrics.json` before deletion; `adk_runs.json` in the results package):

Model	benchmark	att.	LLM calls	tool errors	refine cycles	max ctx [k]	exit	stage / result
gemini-3.1-pro-preview	Heavy N	1	150	51	50	98	max_llm_calls	model_written; fail (model)
gemini-3.1-pro-preview	Heavy N	2	67	18	17	45	completed (gave up)	model_written; fail (model)
gemini-3.1-pro-preview	Heavy N	3	102	44	43	64	completed	figure; success
gemini-3.1-pro-preview	U(1)’ scan	1	150	39	39	143	max_llm_calls	events; fail (generation)
gemini-3.1-pro-preview	U(1)’ scan	2	150	25	23	219	max_llm_calls	compiled; fail (generation)
gemini-3.1-pro-preview	U(1)’ scan	3	28	2	2	39	completed	figure; success
gemini-3.1-pro-preview	ALP EFT	1	150	30	30	92	completed (at the cap)	figure; success
gemini-3.1-pro-preview	ALP EFT	2	150	38	36	91	max_llm_calls	ufo; fail (model)
gemini-3.1-pro-preview	ALP EFT	3	107	28	28	96	completed	figure; success
gemini-3.1-pro-preview	General Z’	1	150	9	9	80	max_llm_calls	figure (empty plot); fail (analysis)
gemini-3.1-pro-preview	General Z’	2	94	6	6	92	completed	figure (curves inverted); fail (model)
gemini-3.1-pro-preview	General Z’	3	20	1	1	31	completed	figure; success
gemini-3.1-pro-preview	KK graviton	1	37	7	7	45	completed	figure; success
gemini-3.1-pro-preview	KK graviton	2	126	30	30	75	completed	figure; success
gemini-3.1-pro-preview	KK graviton	3	150	52	50	70	max_llm_calls	events; fail (generation)
gemini-3.1-pro-preview	U1 LQ at MuC, η	1	150	108	107	379	max_llm_calls	events; fail (generation)
gemini-3.1-pro-preview	U1 LQ at MuC, η	2	150	78	77	182	max_llm_calls	figure (identical panels); fail (model)
gemini-3.1-pro-preview	U1 LQ at MuC, η	3	150	121	119	230	max_llm_calls	events; fail (generation)
gemini-3.1-pro-preview	U1 LQ at MuC, reach	1	108	41	41	195	completed	figure (empty plot); fail (analysis)
gemini-3.1-pro-preview	U1 LQ at MuC, reach	2	150	58	57	149	max_llm_calls	events; fail (generation)
gemini-3.1-pro-preview	U1 LQ at MuC, reach	3	120	41	40	106	completed	figure; success
gemini-3.1-pro-preview	Scalar LQ m_{e_j} (Delphes)	1	300	81	81	199	max_llm_calls (cap 300)	events; fail (generation)
gemini-3.1-pro-preview	U1 LQ mono- τ (Delphes)	1	300	22	20	139	max_llm_calls (cap 300)	ufo; fail (generation)
gpt-5.5	Heavy N	1	74	12	12	110	completed	figure; success
gpt-5.5	U(1)’ scan	1	26	4	4	68	completed	figure; success
gpt-5.5	ALP EFT	1	19	2	2	58	completed	figure; success
gpt-5.5	General Z’	1	26	1	1	50	completed	figure; success
gpt-5.5	KK graviton	1	34	4	4	69	completed	figure; success
gpt-5.5	U1 LQ at MuC, η	1	32	2	2	72	completed	figure; success
gpt-5.5	U1 LQ at MuC, reach	1	34	4	4	72	completed	figure; success
gpt-5.5	Scalar LQ m_{e_j} (Delphes)	1	67	17	17	115	completed	figure (parton level only); fail (generation)
gpt-5.5	U1 LQ mono- τ (Delphes)	1	42	9	8	66	completed	figure (LHE-level selection); fail (generation)

7 Findings

- Same model, different harness: 8/21 vs 7/7.** Gemini 3.1Pro inside Gemini CLI (skills, shell, file tools, automatic context management) reproduced all seven figure benchmarks at 0.63 h and 72 tool calls per run; inside the ADK single loop it succeeded in 8 of 21 attempts, averaging 117 LLM calls and 39 tool errors, with 10 attempts ending at the call cap. Of the 13 failures, 10 are budget exhaustion inside event-generation or model-building error loops (contexts of 379k/182k/230k tokens in the three MuC η attempts), 2 are the agent giving up, 1 is a wrong coupling assignment.
- The strongest model masks the structural penalty on the figure benchmarks.** GPT-5.5 in the identical ADK loop went 7/7 with 35 calls and 4 tool errors per run. The single-loop design hurts the previous-generation model, not the frontier one, on tasks of this size.
- The Delphes benchmarks separate the harnesses again.** They need a lepton→photon LHE rewrite, 100k-event Pythia8+Delphes samples split to fit the 10 GB job limit, and hours of polling. Coding agents wrote the scripts and re-launched (Codex GPT-5.5: Scalar LQ 4.4 h, mono- τ 2.6 h; GPT-5.3-Codex: mono- τ 2.6 h with the paper’s exclusion): 4/6. Both ADK models failed all four attempts: Gemini never got past compilation or a launch loop within 300 calls; GPT-5.5 replaced the detector simulation by parton-level approximations when its fixed tool set did not cover the obstacle (footnotes h). ADK: 0/4.
- Failure kinds differ.** ADK failures are loops (generation 8, model 5, analysis 2 across 23 attempts); the Claude Code and Codex failures are single physics errors (the MuC η leptoquark coupling not taking effect in Opus 4.6 attempt 2 and in GPT-5.3-Codex; an empty MuC-reach contour in GPT-5.3-Codex) or infrastructure (mono- τ early stop, Scalar LQ disk limit, PDF installer).
- Skills are portable across coding agents.** The same `.agents/skills` drove the full pipeline (validate → UFO → compile → launch → analysis, with step sidecars) in Claude Code, Codex and Gemini CLI; Gemini CLI activated 6–8 skills per run without custom sub-agents.
- Deviations are run-level analysis choices, not model- or prompt-level constants.** The U(1)’ 3 TeV contour tip is 0.48 in Opus 4.6 ($\times 3$), Opus 5, GPT-5.3-Codex and GPT-5.5/ADK, 0.55 in Gemini CLI and the paper’s 0.65 only in Codex GPT-5.5; the mono- τ bound is 20–35 % weak in the Opus 4.6 continuation and Codex GPT-5.5 but matches the paper in GPT-5.3-Codex (0.88 vs 0.8 at 0.8 TeV).
- Caveats.** (i) Model and harness are confounded in column C; the two controls bound the effect but an Anthropic model inside the ADK loop, or Gemini in ADK through the native client, would tighten it: ADK’s Gemini went through an OpenAI-compatible relay that drops Gemini 3 thought signatures, Gemini CLI through the native API. (ii) The ADK agent received `run_python/run_shell` and a call cap of 150/300; Claude Code and Codex have no cap but used far fewer calls. (iii) The KK-graviton ADK successes have on-pole cross sections two orders of magnitude above the reference (no width smearing); they pass the qualitative criterion with a footnote.

8 Incidents and fixes during the campaign

- Headless claude -p kills background sub-agents 600s after the main agent ends its turn: the harness sets CLAUDE_CODE_PRINT_BG_WAIT_CEILING_MS to 0 (2026-09-16).
- mono- τ attempt 1: the orchestrator ended its turn “waiting for downloads”; the orchestrator skill now forbids ending a turn to wait (commit e7c5b61). Scalar LQ attempt 1: 27 GB Delphes ROOT against 10 GB job space; the madgraph-simulator skill documents the limit and the post-run cleanup (61800a9).
- Copied OAuth tokens expire when any session refreshes: 4 of 7 attempt-3 runs died at 38–40 min on 2026-09-24; the harness now symlinks .credentials.json (9689e8b) and the three figure-less runs were relaunched.
- The assistive judge once emitted “;”-separated JSON; judge.sh repairs it (2026-09-24).
- The Magnus job list is shared and never purged (833 jobs): a GPT-5.3-Codex run reproduced Heavy N in 7 min from the 2026-09-16 job logs. Quarantined and re-run; benchmark_rules.md appended to every prompt; job_provenance.py verified all other 52 runs of 2026-09-26 (c565067, 36b2588, d3211ef).
- madgraph-launch -pdf cannot install LHAPDF sets on zhustation (lhpdf-config not on PATH); the MG5-internal pdlabel/lhaid route works and is what the successful Scalar LQ runs used.
- Home disk at 97 %: 2026-09-26 sandboxes lived on /playpen1 and were deleted after packaging (15 GB).

9 Open items

1. Opus 4.6 $\times 3$ on Scalar LQ and mono- τ (footnotes a, b of the claims audit; 1.5–6 h and >100 USD each at xhigh, --effort high recommended).
2. $\times 3$ for Opus 5 and GPT-5.3-Codex (one attempt each so far).
3. A stricter model/harness separation: an Anthropic model inside the ADK loop, or Gemini in ADK through the native client.

A Every attempt of the campaign

date	benchmark	column	att.	wall [h]	jobs	tok in [M]	USD	result
2026-09-16	Heavy N	Opus 4.6 / Claude Code	1	0.40	7	3.2	4.2	ok
2026-09-16	U(1)' scan	Opus 4.6 / Claude Code	1	1.05	17	5.2	9.2	ok*
2026-09-16	U(1)' scan	Opus 5 / Claude Code	1	0.88	7	9.1	14.3	ok*
2026-09-16	ALP EFT	Opus 4.6 / Claude Code	1	0.86	6	36.9	22.2	ok
2026-09-16	ALP EFT	Opus 5 / Claude Code	1	0.87	5	5.7	9.1	ok
2026-09-16	ALP EFT	Opus 4.8 / Claude Code	1	1.05	9	11.0	12.6	ok
2026-09-16	ALP EFT	Sonnet 5 / Claude Code	1	1.38	14	18.2	7.5	ok*
2026-09-16	General Z'	Opus 4.6 / Claude Code	1	0.66	7	19.4	13.3	ok
2026-09-16	KK graviton	Opus 4.6 / Claude Code	1	1.35	16	4.2	6.4	ok*
2026-09-16	U1 LQ at MuC, η	Opus 4.6 / Claude Code	1	0.55	14	1.3	5.9	ok*
2026-09-16	U1 LQ at MuC, reach	Opus 4.6 / Claude Code	1	1.41	25	27.3	20.6	ok*
2026-09-16	Scalar LQ m_{ej} (Delphes)	Opus 4.6 / Claude Code	1	6.41	42	214.0	-	TBD
2026-09-16	U1 LQ mono- τ (Delphes)	Opus 4.6 / Claude Code	1	4.00	29	181.4	106.0	i
2026-09-24	Heavy N	Opus 4.6 / Claude Code	2	0.37	7	2.7	3.6	ok
2026-09-24	Heavy N	Opus 4.6 / Claude Code	3	1.04	13	5.7	6.7	ok*
2026-09-24	U(1)' scan	Opus 4.6 / Claude Code	2	1.07	17	4.5	7.5	ok*
2026-09-24	U(1)' scan	Opus 4.6 / Claude Code	3	0.65	17	3.6	6.0	ok*
2026-09-24	U(1)' scan	GPT-5.5 / Codex	1	1.03	21	11.1	-	ok*
2026-09-24	ALP EFT	Opus 4.6 / Claude Code	2	1.30	8	5.1	10.5	ok*
2026-09-24	ALP EFT	Opus 4.6 / Claude Code	3	0.87	6	3.2	5.7	ok*
2026-09-24	ALP EFT	GPT-5.5 / Codex	1	0.99	9	13.4	-	ok*
2026-09-24	General Z'	Opus 4.6 / Claude Code	2	0.59	6	2.5	3.9	ok*
2026-09-24	General Z'	Opus 4.6 / Claude Code	3	0.94	9	2.9	4.4	ok*
2026-09-24	KK graviton	Opus 4.6 / Claude Code	2	0.76	7	3.6	6.1	ok*
2026-09-24	KK graviton	Opus 4.6 / Claude Code	3	0.75	5	3.6	6.6	ok*
2026-09-24	U1 LQ at MuC, η	Opus 4.6 / Claude Code	2	0.74	11	21.1	14.5	m
2026-09-24	U1 LQ at MuC, η	Opus 4.6 / Claude Code	3	0.81	14	7.5	9.7	ok*
2026-09-24	U1 LQ at MuC, reach	Opus 4.6 / Claude Code	2	1.38	19	8.2	10.0	ok*
2026-09-24	U1 LQ at MuC, reach	Opus 4.6 / Claude Code	3	0.87	13	4.2	7.4	ok*
2026-09-24	Scalar LQ m_{ej} (Delphes)	GPT-5.5 / Codex	1	4.41	24	75.0	-	ok*
2026-09-24	U1 LQ mono- τ (Delphes)	GPT-5.5 / Codex	1	2.55	20	49.4	-	ok*
2026-09-26	Heavy N	Opus 5 / Claude Code	1	0.71	11	4.6	7.0	ok*
2026-09-26	Heavy N	GPT-5.3-Codex / Codex	1	0.45	10	3.7	-	ok*
2026-09-26	Heavy N	Gemini 3.1 Pro / ADK	1	1.17	24	9.1	-	m
2026-09-26	Heavy N	Gemini 3.1 Pro / ADK	2	0.48	11	2.1	-	m
2026-09-26	Heavy N	Gemini 3.1 Pro / ADK	3	1.63	38	3.9	-	ok*
2026-09-26	Heavy N	Gemini 3.1 Pro / Gemini CLI	1	0.36	5	1.9	-	ok*
2026-09-26	Heavy N	GPT-5.5 / ADK	1	0.74	24	5.4	-	ok*

date	benchmark	column	att.	wall [h]	jobs	tok in [M]	USD	result
2026-09-26	U(1)' scan	GPT-5.3-Codex / Codex	1	1.23	22	6.0	-	ok*
2026-09-26	U(1)' scan	Gemini 3.1 Pro / ADK	1	2.36	40	14.1	-	g
2026-09-26	U(1)' scan	Gemini 3.1 Pro / ADK	2	0.76	20	16.1	-	g
2026-09-26	U(1)' scan	Gemini 3.1 Pro / ADK	3	0.68	18	0.8	-	ok*
2026-09-26	U(1)' scan	Gemini 3.1 Pro / Gemini CLI	1	0.58	12	4.4	-	ok*
2026-09-26	U(1)' scan	GPT-5.5 / ADK	1	0.66	18	1.1	-	ok*
2026-09-26	ALP EFT	GPT-5.3-Codex / Codex	1	0.86	16	7.3	-	ok*
2026-09-26	ALP EFT	Gemini 3.1 Pro / ADK	1	1.14	23	9.1	-	ok*
2026-09-26	ALP EFT	Gemini 3.1 Pro / ADK	2	1.23	32	8.4	-	m
2026-09-26	ALP EFT	Gemini 3.1 Pro / ADK	3	1.04	20	6.9	-	ok*
2026-09-26	ALP EFT	Gemini 3.1 Pro / Gemini CLI	1	0.76	11	2.6	-	ok
2026-09-26	ALP EFT	GPT-5.5 / ADK	1	0.59	5	0.8	-	ok*
2026-09-26	General Z'	Opus 5 / Claude Code	1	1.26	10	6.9	10.7	ok*
2026-09-26	General Z'	GPT-5.3-Codex / Codex	1	1.14	22	6.9	-	ok*
2026-09-26	General Z'	Gemini 3.1 Pro / ADK	1	1.30	18	8.5	-	a
2026-09-26	General Z'	Gemini 3.1 Pro / ADK	2	2.76	38	5.5	-	m
2026-09-26	General Z'	Gemini 3.1 Pro / ADK	3	0.57	4	0.5	-	ok*
2026-09-26	General Z'	Gemini 3.1 Pro / Gemini CLI	1	0.48	10	1.9	-	ok*
2026-09-26	General Z'	GPT-5.5 / ADK	1	0.48	7	0.9	-	ok*
2026-09-26	KK graviton	Opus 5 / Claude Code	1	1.72	12	9.4	13.2	ok*
2026-09-26	KK graviton	GPT-5.3-Codex / Codex	1	0.54	8	5.4	-	ok*
2026-09-26	KK graviton	Gemini 3.1 Pro / ADK	1	0.87	4	1.3	-	ok*
2026-09-26	KK graviton	Gemini 3.1 Pro / ADK	2	1.87	52	6.6	-	ok*
2026-09-26	KK graviton	Gemini 3.1 Pro / ADK	3	2.04	52	6.3	-	g
2026-09-26	KK graviton	Gemini 3.1 Pro / Gemini CLI	1	0.94	14	6.2	-	ok*
2026-09-26	KK graviton	GPT-5.5 / ADK	1	0.44	8	1.6	-	ok*
2026-09-26	U1 LQ at MuC, η	Opus 5 / Claude Code	1	1.18	8	6.5	9.8	ok*
2026-09-26	U1 LQ at MuC, η	GPT-5.3-Codex / Codex	1	0.49	12	3.5	-	m
2026-09-26	U1 LQ at MuC, η	Gemini 3.1 Pro / ADK	1	1.44	21	22.3	-	g
2026-09-26	U1 LQ at MuC, η	Gemini 3.1 Pro / ADK	2	1.53	35	10.4	-	m
2026-09-26	U1 LQ at MuC, η	Gemini 3.1 Pro / ADK	3	1.89	21	17.0	-	g
2026-09-26	U1 LQ at MuC, η	Gemini 3.1 Pro / Gemini CLI	1	0.83	16	5.9	-	ok*
2026-09-26	U1 LQ at MuC, η	GPT-5.5 / ADK	1	0.56	13	1.8	-	ok*
2026-09-26	U1 LQ at MuC, reach	Opus 5 / Claude Code	1	4.27	21	23.1	33.3	ok*
2026-09-26	U1 LQ at MuC, reach	GPT-5.3-Codex / Codex	1	0.96	17	4.0	-	a
2026-09-26	U1 LQ at MuC, reach	Gemini 3.1 Pro / ADK	1	1.33	19	8.3	-	a
2026-09-26	U1 LQ at MuC, reach	Gemini 3.1 Pro / ADK	2	1.92	22	10.2	-	g
2026-09-26	U1 LQ at MuC, reach	Gemini 3.1 Pro / ADK	3	1.56	30	7.1	-	ok*
2026-09-26	U1 LQ at MuC, reach	Gemini 3.1 Pro / Gemini CLI	1	0.46	11	3.0	-	ok*
2026-09-26	U1 LQ at MuC, reach	GPT-5.5 / ADK	1	1.52	19	1.7	-	ok*
2026-09-26	Scalar LQ m_{ej} (Delphes)	GPT-5.3-Codex / Codex	1	0.81	18	11.6	-	g
2026-09-26	Scalar LQ m_{ej} (Delphes)	Gemini 3.1 Pro / ADK	1	2.80	35	31.6	-	g
2026-09-26	Scalar LQ m_{ej} (Delphes)	GPT-5.5 / ADK	1	0.68	24	5.7	-	g
2026-09-26	U1 LQ mono- τ (Delphes)	GPT-5.3-Codex / Codex	1	2.64	8	7.3	-	ok*
2026-09-26	U1 LQ mono- τ (Delphes)	Gemini 3.1 Pro / ADK	1	1.29	20	25.2	-	g
2026-09-26	U1 LQ mono- τ (Delphes)	GPT-5.5 / ADK	1	1.76	12	1.8	-	g

B Quantitative deviations of the successful runs (footnotes)

- 2103.02708 Fig. 4, gpt-5.3-codex / codex:** High-mass normalisation agrees well (SSM $\sim 2.5 \times 10^{-5}$ pb and $\psi \sim 1 \times 10^{-5}$ pb near 5 TeV, comparable to the reference theory curves), but at low mass the produced curves sit roughly a factor 2-4 high ($\psi \sim 9 \times 10^{-2}$ pb at 1 TeV versus $\sim 2 \times 10^{-2}$ pb inferred from the reference), so the slope is slightly shallower than the paper's. The scan also starts near 600 GeV rather than covering the reference's full range down to ~ 250 GeV.
- 9909255 Fig. 2, gpt-5.3-codex / codex:** The produced scan covers 200-1200 GeV and therefore omits the Z pole spike near 91 GeV visible in the reference. Normalisation differs by roughly a factor of 3 at the 600 GeV resonance ($\sim 1.3 \times 10^5$ fb produced vs $\sim 4 \times 10^4$ fb in the reference) and is low by a factor of $\sim 2-3$ in the 400 GeV dip ($\sim 9 \times 10^2$ fb vs $\sim 2 \times 10^3$ fb); the coarse 100 GeV step also under-resolves the resonance widths, so the 600 GeV peak height and the 1100 GeV structure are sampled rather than traced.
- 1308.2209 Fig. 3, claude-opus-5:** Agent's 14 TeV $\sigma(\mu\pm N)$ agrees with the reference red curve to within $\sim 10-20\%$ across the mass range (e.g. ~ 330 fb vs ~ 380 fb at 300 GeV, ~ 2.5 fb vs ~ 3 fb at 1 TeV); the y-axis is quoted as σ [fb] at $|V_{\mu N}| = 1$ rather than the reference's $\sigma/|V_{\mu N}|^2$, which is numerically equivalent.
- 2103.02708 Fig. 3, claude-opus-5:** Absolute normalisation tracks the reference well at high mass (SSM $\approx 2 \times 10^{-5}$ pb near 5 TeV and $Z'_\psi \approx 1.6 \times 10^{-5}$ pb near 4.5 TeV, consistent with the reference crossing points to within $\sim 20-30\%$). The produced figure uses a linear mass axis and extends down to 200 GeV, whereas the reference is logarithmic in mass and truncated at 10^{-1} pb, so the low-mass part of the curves is off-scale in the reference. Two extra 'inclusive' (full Breit-Wigner, no mass window) curves are shown in addition to the resonant ones; they deviate from the resonant curves only above ~ 3 TeV, by up to a factor ~ 4 at 5.5 TeV for the SSM.
- 1701.05379 Fig. 8, gemini-3.1-pro-preview / adk:** The produced y-axis is per-bin event fraction (~ 0.09 at the peak with ~ 20 GeV bins) whereas the reference plots $1/\sigma d(\sigma)/dE_T^{\text{miss}}$ in GeV^{-1} , so the two differ by a bin-width factor of ~ 20 ; after that conversion the peak ($\sim 4.5 \times 10^{-3} \text{ GeV}^{-1}$) sits below the $c1/c2/cW$ curves (~ 0.2) and the tail at 1 TeV ($\sim 1.3 \times 10^{-3}$ per bin, i.e. $\sim 6 \times 10^{-5} \text{ GeV}^{-1}$) falls between the $c1/c2/cW$ ($\sim 3 \times 10^{-4}$) and $c6/c8$ ($\sim 3 \times 10^{-3}$) reference curves. The steeply-falling SM reference curve (which dies out by ~ 250 GeV) is not reproduced, and no per-coefficient breakdown is shown.
- 1308.2209 Fig. 3, gemini-3.1-pro-preview / gemini:** The produced 14 TeV curve is systematically $\sim 10-25\%$ below the reference in the mid-mass range (e.g. $\sim 1.2 \times 10^3$ fb vs $\sim 1.5 \times 10^3$ fb at $m_N = 200$ GeV, ~ 100 fb vs ~ 115 fb at 400 GeV), while the endpoints agree well ($\sim 1.7 \times 10^4$ fb at 100 GeV and ~ 2.5 fb vs ~ 3 fb at 1000 GeV). The produced y-axis is σ in fb rather than the reference's $\sigma/|V_{\mu N}|^2$, i.e. it corresponds to the mixing-independent normalisation $|V_{\mu N}|^2 = 1$.
- 2103.02708 Fig. 4, gemini-3.1-pro-preview / gemini:** Normalisation is not directly comparable since the reference axis is the ratio $(\sigma_B)Z'/(\sigma_B)Z$ scaled by 1928 rather than a cross section in pb; taking the reference curves at face value, the produced Z'_ψ curve sits high by roughly a factor of 3-5 at 1 TeV ($\sim 9 \times 10^{-2}$ pb vs $\sim 2 \times 10^{-2}$) while being low by a similar factor near 5 TeV ($\sim 6 \times 10^{-6}$ pb vs $\sim 2 \times 10^{-5}$), i.e. the produced curves fall somewhat more steeply with mass.
- 1308.2209 Fig. 3, gpt-5.3-codex / codex:** Produced 14 TeV values run slightly below the reference throughout: $\sim 1.5 \times 10^4$ vs $\sim 2 \times 10^4$ fb at 100 GeV, $\sim 1.2 \times 10^3$ vs $\sim 1.5 \times 10^3$ fb at 200 GeV, ~ 100 vs ~ 110 fb at 400 GeV, and ~ 2.5 vs ~ 3 fb at 1000 GeV — a roughly 10-25% normalisation offset, consistent with the reference being normalised as $\sigma/|V_{\mu N}|^2$ while the produced curve is a plain cross section at the chosen mixing.
- 1605.02910 Fig. 1, gemini-3.1-pro-preview / gemini:** The reproduced LHC contour tips sit somewhat higher than the reference: $g'_1 \approx 0.17$ vs ≈ 0.12 (2 TeV), ≈ 0.30 vs ≈ 0.27 (2.5 TeV), and ≈ 0.55 vs ≈ 0.65 (3 TeV, here lower); the lobes are also shifted $\sim 0.05-0.1$ toward negative $\tilde{g}$ at 2 and 2.5 TeV, and the 3 TeV lobe is broader in $\tilde{g}$ (left edge near -0.65 vs -0.75 at the tip with a wider lower crossing spread).

10. **1701.05379 Fig. 8, gpt-5.3-codex / codex:** Only one coupling benchmark is shown, versus the seven curves (c1, c2, c6, c8, cB, cW, SM) of the reference; the produced curve tracks the c2/cW-like family, sitting within roughly a factor of two of them (e.g. $\sim 2.5 \times 10^{-2}$ vs $\sim 3 \times 10^{-2}$ at 200 GeV and $\sim 1.6 \times 10^{-3}$ vs $\sim 8 \times 10^{-4}$ at 600 GeV), and it does not reproduce the much harder c6/c8 spectra or the fast-falling SM and cB tails. The y-axis is labelled 'normalized events' rather than the reference's $(1/\sigma)d\sigma/dE_T$ in GeV^{-1} , so the vertical scale agrees in magnitude but not strictly in units.
11. **9909255 Fig. 2, claude-opus-5:** Scan starts at 200 GeV, so the Z pole near 91 GeV and the low- $\sqrt{s}$ rise visible in the reference are absent. The 100 GeV step size undersamples the resonance: the 600 GeV peak reaches $\sim 3.5 \times 10^4$ fb (reference peak $\sim 4 \times 10^4$ fb) but its true Breit-Wigner width is not resolved, and the second KK resonance near 1100 GeV appears only as a smooth rise to $\sim 2.7 \times 10^4$ fb rather than a distinct peak. The post-resonance minimum sits at $\sim 3 \times 10^3$ fb near 800-900 GeV versus $\sim 2.5 \times 10^3$ fb near 850 GeV for the reference blue curve.
12. **1605.02910 Fig. 1, gpt-5.3-codex / codex:** The LHC contour extent differs modestly from the reference: the maximum $g1'$ reach is ~ 0.17 vs ~ 0.12 at $M_{Z'} = 2$ TeV ($\sim 40\%$ high), ~ 0.25 vs ~ 0.27 at 2.5 TeV (close), and ~ 0.48 vs ~ 0.65 at 3 TeV ($\sim 25\%$ low); the 3 TeV contour's left edge also sits near $\tilde{g} \approx -0.57$ instead of ≈ -0.75 , so the contour is somewhat narrower in $\tilde{g}$ at the highest mass. Lower-edge intercepts on $g1' = 0$ agree well (e.g. ~ -0.07 to $+0.07$ at 2 TeV vs ~ -0.09 to $+0.05$).
13. **9909255 Fig. 2, gemini-3.1-pro-preview / adk:** Sampled on a coarse 100 GeV grid (200-1200 GeV), so the resonances are rendered as single-point spikes rather than resolved Breit-Wigner shapes; the on-peak values ($\sim 1.6 \times 10^6$ fb at 600 GeV and $> 5 \times 10^6$ fb at 1100 GeV) overshoot the reference peak heights ($\sim 4 \times 10^4$ and $\sim 1 \times 10^4$ fb) by one to two orders of magnitude because the grid point sits exactly on the pole with no width smearing. The low-energy region below 200 GeV, including the Z pole and the SM continuum falloff, is not covered, and the off-peak continuum near 200-500 GeV ($\sim 1\text{--}3 \times 10^3$ fb) sits below the reference's $\sim 2 \times 10^3$ fb band.
14. **9909255 Fig. 2, gemini-3.1-pro-preview / gemini:** The scan covers only $\sqrt{s} = 400\text{--}1200$ GeV (points every 100 GeV), so the reference's low-energy region (Z pole near 90 GeV and the falling SM QED tail down to the ~ 300 GeV minimum) and the region up to 1500 GeV are absent; the reference's family of curves for different parameter choices is represented by a single curve. The lowest scan point (≈ 400 GeV) falls below $\sim 4 \times 10^2$ fb and runs off the bottom of the frame, whereas the reference curves are $\sim 1\text{--}2 \times 10^3$ fb there, i.e. the SM-dominated low-energy behaviour is not reproduced. At the resonance the produced peak is $\approx 3.3 \times 10^4$ fb versus $\approx 4 \times 10^4$ fb in the reference, and the 800 GeV dip ($\approx 2.7 \times 10^3$ fb) and 1200 GeV value ($\approx 2.6 \times 10^4$ fb) agree with the reference blue curve to within roughly 10-20%; the coarse 100 GeV grid also means the resonance peak height is only sampled, not resolved.
15. **2104.05720 Fig. 11, claude-opus-5:** In the left panel the produced $m_{LQ} = 1$ TeV curve peaks slightly further forward ($\eta \sim 1.3\text{--}1.5$ vs $\sim 1.1\text{--}1.4$ in the reference) and stays $\sim 30\text{--}60\%$ higher in the tail (~ 0.04 vs $\sim 0.02\text{--}0.03$ at $\eta \sim 2.3\text{--}2.5$); the reference plots η_{μ} whereas the produced figure labels the axis simply η .
16. **2104.05720 Fig. 11, gpt-5.3-codex / codex:** SM baseline reproduces the reference well (first bin ~ 0.075 , peak $\sim 0.150\text{--}0.155$ at $\eta \sim 0.2$, falling to ~ 0.01 at $\eta \sim 2.4$). Deviation is in the BSM curves: reference $m_{LQ}=1$ TeV, $\beta_{L32}=1$ peaks at ~ 0.13 near $\eta \sim 1.4$ and starts at ~ 0.02 in the first bin, whereas the produced curve peaks at ~ 0.151 at $\eta \sim 0.2$ and starts at ~ 0.075 , i.e. matches the SM within $\sim 1\text{--}2\%$ everywhere; the $m_{LQ}=10$ TeV curve agrees with the reference (slightly above SM in the first two bins, ~ 0.155 vs ~ 0.150).
17. **1308.2209 Fig. 3, gpt-5.5 / adk:** At $\sqrt{s} = 14$ TeV the produced cross sections track the reference $pp \rightarrow N\bar{N} \pm$ curve closely over the full mass range (e.g. $\sim 1.5 \times 10^4$ fb vs $\sim 2 \times 10^4$ fb at 100 GeV, $\sim 1.2 \times 10^3$ vs $\sim 1.5 \times 10^3$ fb at 200 GeV, ~ 105 vs ~ 110 fb at 400 GeV, ~ 2.6 vs ~ 3 fb at 1000 GeV), i.e. agreement within roughly 10-25%, with the produced values sitting slightly low at the lightest masses. The reference normalises to $\sigma/[V_{\mu N}]^2$ while the produced figure fixes $|V_{\mu N}| = 1$, which is equivalent.
18. **1701.05379 Fig. 8, gpt-5.5 / adk:** The produced histogram is normalized to unit total area per bin (dimensionless 'normalized events') rather than the reference's $1/\sigma d\sigma/dMET$ in GeV^{-1} , so absolute y-values are not directly comparable. The shape most closely tracks the reference $c_{1/c} 2/c_W$ curves; the harder c_6/c_8 curves (which peak around 150-250 GeV and fall far more slowly) are not represented, and the produced tail falls somewhat more slowly than the SM/ c_B curves.
19. **2104.05720 Fig. 11, gemini-3.1-pro-preview / gemini:** In the $\beta_{L32} = 1.0$ panel the reproduced $m_{LQ} = 1$ TeV curve peaks slightly earlier ($|\eta| \sim 1.1\text{--}1.3$ at ~ 0.118) than the reference ($\sim 1.3\text{--}1.5$ at ~ 0.13) and stays higher in the tail (~ 0.04 vs ~ 0.02 near $|\eta| \sim 2.4$). In the $\beta_{L32} = 0.1$ panel the reproduced 1 TeV curve is slightly more peaked at low $|\eta|$ (~ 0.17 vs ~ 0.175 leading bin, with a marginally deeper dip near $|\eta| \sim 1.2$). The reference plots signed η_{μ} while the reproduction uses $|\eta|$, which is equivalent given the symmetric range shown.
20. **1811.07920 Fig. 3, gpt-5.3-codex / codex:** The produced plot omits the reference's auxiliary grey EFT dash-dotted line and the $150 \text{ fb}^{-1} / 3 \text{ ab}^{-1}$ projection curves, and carries no in-panel RH/LH/EFT text labels. The black 2σ contour nearly coincides with the upper edge of the red RH band over the whole mass range, whereas in the reference the grey-region boundary sits slightly above the RH band at low mass; endpoint values (≈ 0.88 at 0.8 TeV, ≈ 3.75 at 5 TeV for the contour; $\approx 0.3\text{--}1.95$ for LH) are within a few percent of the reference.
21. **2104.05720 Fig. 12, claude-opus-5:** Limits are somewhat weaker than the reference at large mass: the 3 TeV 5-sigma curve reaches $\beta_{L32} \sim 2$ at $m_{LQ} = 55$ TeV versus ~ 1 in the reference, and the 14 TeV 5-sigma curve ~ 0.45 versus ~ 0.3 . At low mass the agreement is good for the discovery curves (~ 0.05 and ~ 0.022 at 1 TeV vs ~ 0.04 and ~ 0.02), but the 14 TeV 95% CL exclusion is notably less constraining at 1 TeV (~ 0.019 vs ~ 0.006 in the reference), so the reproduced exclusion/discovery separation at 14 TeV is narrower than in the paper.
22. **1701.05379 Fig. 8, gemini-3.1-pro-preview / adk:** Only one curve is drawn, so no SM-vs-operator comparison is possible; its tail ($\sim 1 \times 10^{-3}$ at 1000 GeV, falling ~ 2 decades from the peak) is much harder than the reference SM curve (which dies below 1×10^{-4} by ~ 250 GeV) and closest to the c_6 curve, while the reference c_8 curve is roughly a factor of ~ 5 higher at 1 TeV. The y-axis is plotted as per-bin 'normalized events' rather than the reference's $(1/\sigma)d\sigma/dE_T$ in GeV^{-1} , so absolute vertical values differ by the bin width (peak ~ 0.12 vs $\sim 0.2\text{--}0.5$ in the reference).
23. **2103.02708 Fig. 4, gemini-3.1-pro-preview / adk:** The reference y-axis is the scaled ratio $(\sigma_B)Z'/(\sigma_B)Z \times 1928$ rather than an absolute cross section in pb, so the normalisations are not directly comparable. More notably, the relative ordering of the two benchmarks is inverted: in the reference the Z'_{SSM} (dotted) curve lies above Z'_{ψ} over the whole mass range ($\sigma_{SSM} \approx 3 \times \sigma_{\psi}$), while the produced figure places Z'_{extSSM} roughly a factor 2-3 below Z'_{ψ} . The produced Z'_{ψ} curve also reaches $\sim 5 \times 10^{-6}$ pb only near 5.1 TeV, i.e. the two curves cross the low-cross-section region a few hundred GeV apart from the reference.
24. **2104.05720 Fig. 12, gemini-3.1-pro-preview / gemini:** Produced contours sit $\sim 20\text{--}30\%$ above the reference at high mass (3 TeV discovery reaches ~ 1.8 vs ~ 1.1 at 50 TeV; 14 TeV discovery ~ 0.5 vs ~ 0.3). The 14 TeV 95% CL curve is too close to its 5-sigma counterpart at low mass (~ 0.016 at 1 TeV vs ~ 0.006 in the reference), so the exclusion/discovery gap is compressed. Minor non-physical kinks appear near $m_{LQ} = 1.5\text{--}2$ TeV in the 14 TeV curves, likely from a coarse mass grid.
25. **9909255 Fig. 2, gemini-3.1-pro-preview / adk:** The scan uses ~ 100 GeV steps, so the resonances appear as triangular spikes rather than resolved Breit-Wigner peaks: the 600 GeV point reaches $> 4 \times 10^6$ fb and the 1100 GeV point $\approx 2 \times 10^6$ fb, roughly two orders of magnitude above the reference peak heights ($\sim 4 \times 10^4$ and $\sim 10^4$ fb), while the inter-resonance minimum near 800 GeV falls below the plotted range instead of the reference's $\sim 1.5 \times 10^2$ fb floor. The plotted range starts near 420 GeV, so the low- $\sqrt{s}$ region including the Z pole and the SM continuum fall-off ($\sqrt{s} \leq 400$ GeV in the reference) is not covered, and the curve stops at 1200 GeV rather than 1500 GeV.
26. **1605.02910 Fig. 1, gpt-5.5 / adk:** Panel (c) ($M_{Z'} = 3$ TeV) reproduces a somewhat smaller contour than the reference: the tip reaches $g1' \approx 0.48$ at $\tilde{g} \approx -0.55$ versus ≈ 0.65 at $\tilde{g} \approx -0.72$ in the paper ($\sim 25\text{--}30\%$ low in $g1'$ and in leftward extent), although the right-hand $g1' = 0$ intercept ($\tilde{g} \approx 0.18$) agrees. Panels (a) and (b) agree to within a few percent (tip $\approx (-0.15, 0.12)$ vs $(-0.13, 0.12)$; $\approx (-0.28, 0.245)$ vs $(-0.30, 0.265)$).
27. **2103.02708 Fig. 4, gpt-5.5 / adk:** Normalisation runs somewhat high relative to the reference theory curves: the produced Z'_{SSM} cross section is $\sim 1 \times 10^{-4}$ pb at 4.3 TeV versus $\sim 3 \times 10^{-5}$ pb in the reference (factor ~ 3), while the Z'_{ψ} curve agrees to within roughly a factor of 1.5 ($\sim 4 \times 10^{-5}$ pb vs $\sim 3 \times 10^{-5}$ pb near 4 TeV). The produced curve stops at the point where it leaves the plotted y-range (~ 5.2 TeV for Z'_{ψ}) rather than extending to 5.5 TeV.
28. **1308.2209 Fig. 3, gemini-3.1-pro-preview / adk:** The produced 14 TeV curve sits $\sim 20\text{--}30\%$ below the reference throughout: $\sim 1.6 \times 10^4$ fb vs $\sim 2 \times 10^4$ fb at $m_N = 100$ GeV, ~ 300 fb vs ~ 400 fb at 300 GeV, and ~ 2.5 fb vs ~ 3 fb at 1000 GeV. The reference normalises to $\sigma/[V_{\mu N}]^2$ while the produced figure plots σ in fb at an implicit mixing value, so the offset may be partly a normalisation convention rather than a physics difference.
29. **2104.05720 Fig. 11, gemini-3.1-pro-preview / adk:** The SM-background (and 10 TeV) histogram exceeds the 0.20 y-axis limit and is clipped over $0.1 < |\eta| < 1.1$, while the reference SM distribution peaks at ~ 0.15 near $\eta \sim 0.2$ and falls monotonically; the 1 TeV signal peaks at $|\eta| \sim 1.0$ instead of $\sim 1.3\text{--}1.5$, and its high- $|\eta|$ tail (~ 0.06 at 2.5) is about 4x the reference value.
30. **1605.02910 Fig. 1, gemini-3.1-pro-preview / adk:** The 3 TeV LHC contour is noticeably smaller than the reference: its apex reaches

- $g_1' \approx 0.47$ at $\tilde{g} \approx -0.55$ versus ≈ 0.65 at $\tilde{g} \approx -0.65$, and its left-hand zero crossing sits at $\tilde{g} \approx -0.2$ instead of ≈ -0.25 . The 2.5 TeV contour is also slightly compressed (apex $g_1' \approx 0.24$ vs ≈ 0.27 , and it is thinner in the $\tilde{g}$ direction near the apex), while the 2 TeV panel agrees to within $\sim 10\%$ (apex $g_1' \approx 0.13$ vs ≈ 0.12). The x-axis label is mislabelled 'ildeg' rather than $\tilde{g}$.
31. **9909255 Fig. 2, gpt-5.5 / adk:** The scan covers 200-1200 GeV in 100 GeV steps, so the Z pole at 91 GeV and the region above 1200 GeV present in the reference are not sampled, and the coarse binning under-resolves the narrow 600 GeV resonance (peak reaches $\sim 3 \times 10^4$ fb vs $\sim 4 \times 10^4$ fb for the reference blue curve). Only a single coupling choice is shown, whereas the reference overlays six curves; the near-1100 GeV second KK resonance is therefore only visible as a rising edge rather than a resolved peak.
 32. **2104.05720 Fig. 11, gpt-5.5 / adk:** In the beta $L^{\wedge}32 = 1.0$ panel the $m_{LQ} = 1$ TeV curve falls off more slowly at large pseudorapidity than in the reference: the produced distribution is ~ 0.04 in the last bin ($|\eta| \sim 2.3-2.5$) versus $\sim 0.01-0.02$ in the reference, with a correspondingly broader peak ($\sim 1.1-1.5$ at 0.117 versus a sharper peak near 1.4-1.5 at ~ 0.13). The axis is labelled $|\eta_b|$ (b-quark) whereas the reference labels it η_{μ} , though the plotted range and content are the same.
 33. **2103.02708 Fig. 4, gemini-3.1-pro-preview / adk:** Normalisation runs roughly 2-4x above the reference theory curves (e.g. Z'_{ψ} at 2 TeV: $\sim 4\text{-}3$ pb produced vs $\sim 1\text{-}3$ in the reference, where the y-axis is $(\sigma \cdot B)_{Z'}/(\sigma \cdot B)_{Z \times 1928}$); the high-mass tails converge better (Z'_{SSM} at ~ 4.7 TeV: $\sim 7\text{-}5$ vs $\sim 1\text{-}4$). The dotted curve is labelled Z'_{extSSM} rather than the reference's Z'_{SSM} .
 34. **2104.05720 Fig. 12, gpt-5.5 / adk:** The 3 TeV curves agree closely with the reference ($\beta_{L^{\wedge}32} \approx 0.04-0.05$ at $m_{LQ} = 1$ TeV, $\approx 1-2$ at 50 TeV). The 14 TeV exclusion curve is roughly 2-3x weaker than the reference at low mass (≈ 0.016 vs ≈ 0.006 at $m_{LQ} = 1$ TeV) and converges to within $\sim 20-30\%$ above ~ 10 TeV; the produced 14 TeV curves also show a small non-monotonic kink near $m_{LQ} \approx 1.5-2$ TeV (likely statistical noise in the scan) that is absent from the smooth reference contours.
 35. **2104.05720 Fig. 12, gemini-3.1-pro-preview / adk:** Sensitivity is $\sim 20-60\%$ weaker than the reference at low mass (3 TeV 5 σ : 0.048 vs ~ 0.04 at $m_{LQ} = 1$ TeV) and roughly a factor 1.5-2 weaker at the high-mass end (3 TeV 5 σ reaches ~ 2 vs ~ 1.2 near 50 TeV). The 14 TeV 95% CL curve is the largest outlier: ~ 0.017 at 1 TeV versus ~ 0.006 in the reference, so the separation between the 14 TeV 5 σ and 95% CL contours is much narrower than in the paper. The curves also show visible kinks (coarse mass grid) where the reference is smooth, and the reference's non-monotonic dip near 1.5-2 TeV in the 14 TeV curves is an artefact not present in the paper.
 36. **2005.06475 Fig. 2, gpt-5.5 / adk:** The reproduced signal peak reaches $\sim 1.8 \times 10^2$ events/bin/100 fb $^{-1}$ versus ~ 4 in the reference black LQ curve, while the low-mass end (~ 0.04 at 1 TeV) is close to the reference (~ 0.02), i.e. the peak-to-shoulder contrast is roughly an order of magnitude larger, consistent with the reference curve being drawn after the full event selection (and possibly a different branching-ratio/normalisation convention). The peak position (~ 3 TeV) and the shape of the falling high-mass tail agree.
 37. **1811.07920 Fig. 3, gpt-5.5 / adk:** The reproduced 2σ exclusion boundary sits somewhat above the reference one at high mass (≈ 2.45 at 5 TeV vs ≈ 3.5 for the reference's boundary between the grey and white regions), so the excluded region is smaller; the reference exclusion edge tracks near the RH band up to ~ 1.5 TeV and then runs parallel above it, whereas the reproduced contour bends below the RH band above ~ 1.3 TeV, crossing it. Band widths are slightly wider than the reference, and the produced contour shows mild kinking from a coarse mass grid.
 38. **2005.06475 Fig. 2, claude-opus-4-6:** TBD (handed over). Stopped by the operator after 6.4 h without a figure: model, validation and the 100k-event LUXlep parton sample were done; two Pythia8+Delphes launches failed at upload with 'No space left on device' (27 GB Delphes ROOT against the job's 10 GB packing space); a third launch with post-run cleanup was running when stopped.
 39. **1605.02910 Fig. 1, claude-opus-4-6:** The 2 TeV and 2.5 TeV contours match the reference to within about 10% (tips at $(-0.155, 0.13)$ vs $(-0.14, 0.12)$ and $(-0.28, 0.245)$ vs $(-0.30, 0.265)$). At 3 TeV the base on $g_1'=0$ matches but the tip is at $(-0.55, 0.48)$ vs $(-0.74, 0.65)$, about 25% smaller in both g_1' and $|\eta|$. An Opus 5 run and the Opus 4.6 attempt 3 gave the same 3 TeV tip, whereas the Codex/gpt-5.5 run of 2026-09-24 reproduced the paper's tip $(-0.72, 0.65)$: the shortfall is an analysis choice made in the Claude runs, not a limitation of the prompt (earlier wording 'systematic to the prompt' is withdrawn).
 40. **1811.07920 Fig. 3, claude-opus-4-6:** Attempt 1 generated all 18 Delphes runs (4.0 h, 106 USD); four 9-run launches were lost to the 10 GB job limit before the agent split them), then the orchestrator ended its turn 'waiting for event file downloads' and the headless session terminated without the analysis stage. A continuation session in the same workspace produced the figure in 1.7 h / 29.5 USD with the reference's shape but a 20-35% weaker exclusion. Counted as a failure (infrastructure).
 41. **9909255 Fig. 2, claude-opus-4-6:** The 11-point sqrt(s) grid (200-1200 GeV, 100 GeV steps) and the single coupling point are what the benchmark prompt specifies, so the reference's Z-pole region, its 1.2-1.5 TeV range and its other coupling curves are outside the task. The grid under-resolves the 600 GeV KK resonance: produced peak about $1.5\text{e}4$ fb against about $3\text{e}4$ fb in the reference; off-peak values follow the reference curve.
 42. **2104.05720 Fig. 11, claude-opus-4-6:** The orchestrator was not triggered: the main session ran the pipeline itself (0 sub-agent calls); the result is correct.
 43. **2104.05720 Fig. 12, claude-opus-4-6:** The produced limits are weaker than the reference throughout: the 14 TeV 95% CL curve at $m_{LQ} = 1$ TeV is about 0.015 against about 0.006 (roughly 2.5x weaker); at 70 TeV the 14 TeV curves are about 0.25 (exclusion) and 0.4 (discovery) against about 0.18 and 0.25. The 3 TeV curves agree within 20-30% at low mass but are about 1.6x weaker at 70 TeV for the exclusion.
 44. **1701.05379 Fig. 8, claude-sonnet-5:** Cross section 0.04909 pb instead of the 0.04135 pb obtained by Opus 5 and Opus 4.8, from looser generation-level η cuts (5 instead of 2.5); the normalization removes the difference.
 45. **1605.02910 Fig. 1, claude-opus-5:** At $M_{Z'} = 3$ TeV the produced contour tip is at g_1' about 0.48 (gt about -0.55) against about 0.65 (gt about -0.73) in the reference, about 25% lower; the $g_1' = 0$ intercepts (about $+0.17$) agree. 2 TeV tip about 0.135 against 0.12; 2.5 TeV tip about 0.245 against 0.265. Same 3 TeV contour as the Opus 4.6 runs; see the 1605.02910 Opus 4.6 attempt-1 note.
 46. **1308.2209 Fig. 3, claude-opus-4-6:** Normalisation convention matches ($|V_{\mu N}| = 1$ vs the reference's σ/V_{IN}^2). Point-by-point the 14 TeV curve sits $\sim 20-30\%$ below the reference at the low-mass end ($1.4\text{e}4$ vs $\sim 2\text{e}4$ fb at $m_N = 100$ GeV, $1.2\text{e}3$ vs $\sim 1.4\text{e}3$ fb at 200 GeV) and agrees to within $\sim 10-20\%$ above 400 GeV; the produced curve uses CTEQ6L, and no PDF/scale uncertainty band is shown.
 47. **1605.02910 Fig. 1, claude-opus-4-6:** Contour tips are systematically slightly lower/narrower than the reference: panel (a) peaks at $g_1' \approx 0.13$ at $\tilde{g} \approx -0.15$ vs ≈ 0.12 at $\tilde{g} \approx -0.13$ (close); panel (b) peaks at ≈ 0.24 vs ≈ 0.27 and its left tail closes near $\tilde{g} \approx -0.3$ vs ≈ -0.32 ; panel (c) is the largest deviation, peaking at $g_1' \approx 0.47$ at $\tilde{g} \approx -0.55$ versus ≈ 0.65 at $\tilde{g} \approx -0.7$ in the reference, i.e. the 3 TeV exclusion region is roughly 25-30% smaller in the vertical extent. The $\tilde{g}$ -axis zero crossings ($\approx 0.05, \approx 0.1, \approx 0.2$ for the three masses) agree well with the reference.
 48. **1605.02910 Fig. 1, claude-opus-4-6:** The contour tips are close for the two lighter masses ($g_1' \approx 0.13$ vs 0.117 at 2 TeV; 0.24 vs 0.265 at 2.5 TeV) but the 3 TeV panel is noticeably smaller: peak $g_1' \approx 0.47$ vs ≈ 0.65 in the reference, and the left edge reaches $\tilde{g} \approx -0.57$ instead of ≈ -0.78 , i.e. a $\sim 25-30\%$ under-coverage of the excluded region at the highest mass. Zero-crossings on the $\tilde{g}$ axis agree to within ~ 0.05 in all panels. The session ended with 'OAuth session expired and could not be refreshed' after the figure and the step-4 sidebar were written (only the execution summary is missing); the figure is judged as produced.
 49. **1605.02910 Fig. 1, gpt-5.5 / codex:** Panel (a) and (c) contours match the reference almost exactly (tip at $g_1' \approx 0.12$ at $\tilde{g} \approx -0.13$, and $g_1' \approx 0.65$ at $\tilde{g} \approx -0.72$ respectively); panel (b) reproduces the tip height ($g_1' \approx 0.26$) and right-hand crossing ($\tilde{g} \approx +0.09$) but the lobe is somewhat narrower/more pinched than the reference, whose contour is fuller in the $0.1 < g_1' < 0.25$ region.
 50. **1701.05379 Fig. 8, claude-opus-4-6:** The reference shows seven overlaid curves ($c_1, c_2, c_6, c_8, c_B, c_W$ and SM) while the reproduction shows a single distribution; its shape tracks the $c_1/c_2/c_W$ -type curves (peak 40-80 GeV, $\sim 3\text{-}4$ at 1 TeV) rather than the much softer SM curve, which dies out by ~ 250 GeV. The y-axis is a per-bin fraction ('normalized events') rather than the reference's density ($1/\sigma d\sigma/dE_T^{\text{miss}}$) in GeV^{-1} , so the absolute level differs by roughly the 20 GeV bin width (peak ~ 0.18 per bin vs ~ 0.2 GeV^{-1}); the intermediate region near 200 GeV sits about a factor of 2 above the c_1 curve and below c_2 .
 51. **1701.05379 Fig. 8, claude-opus-4-6:** The produced histogram plots normalized events per bin rather than the reference's $(1/\sigma)d\sigma/dE_T$ density in GeV^{-1} , so absolute ordinate values differ by the bin width (peak 0.18 per 25 GeV bin vs $\sim 0.2-0.25$ GeV^{-1} in the reference). The tail at 1 TeV lands at $\sim 3 \times 10^{-4}$, consistent with the reference's $c_1/c_2/c_W$ curves but $\sim 20\text{x}$ below its c_6/c_8 curves; only one benchmark coefficient is shown, and no SM curve is overlaid.
 52. **1701.05379 Fig. 8, gpt-5.5 / codex:** The produced histogram falls more slowly than the reference's $c_1/c_2/c_W/c_B$ curves and faster than c_6/c_8 : at 1000 GeV it sits at $\sim 2\text{-}4$, versus $\sim 3\text{-}4$ ($c_1/c_2/c_W$), $\sim 3\text{-}3$ (c_6) and $\sim 6\text{-}3$ (c_8) in the reference. The peak height is ~ 0.18 vs $\sim 0.2-0.25$ for most reference curves, and the low- E_T rise is smoother (finer binning) than the reference's sharp SM-like spike at ~ 50 GeV. The produced curve is a per-bin normalized fraction rather than the reference's $1/\sigma d\sigma/dE_T$ density, so absolute y-values differ by the bin width.

53. **1811.07920 Fig. 3, gpt-5.5 / codex:** The LH and RH R_D(*) bands agree closely with the reference (LH from ~ 0.3 at 0.8 TeV to ~ 1.95 at 5 TeV; RH from ~ 0.65 to ~ 4). The exclusion boundary is somewhat weaker/steeper than the reference: it starts near 1.07 at 0.8 TeV (vs ~ 0.8 in the reference, where the boundary essentially tracks the RH band at low mass) and reaches 4.0 near $M_{U1} \approx 3.9$ TeV, whereas the reference excluded region hugs the RH band up to ~ 5 TeV. The reference's EFT-validity line and the 150 fb $^{-1}$ and 3 ab $^{-1}$ projection curves are not drawn.
54. **2005.06475 Fig. 2, gpt-5.5 / codex:** The signal peak sits at ~ 1.5 events/bin versus ~ 3 in the reference (roughly a factor 2 lower normalisation), and the produced distribution terminates at ~ 3.8 TeV whereas the reference LQ curve extends to 5 TeV with a low tail; the low-mass shoulder is also ~ 2 -5x below the reference near 1-2 TeV, and the x-axis is drawn linearly rather than logarithmically.
55. **2103.02708 Fig. 4, claude-opus-4-6:** The produced curves sit high relative to the reference theory lines: $\sigma_B(Z'_{SSM}) \approx 2.6 \times 10^{-1}$ pb at 1 TeV versus ~ 2 - 3×10^{-2} pb in the reference (factor ~ 10), narrowing to $\sim 4 \times 10^{-5}$ pb versus $\sim 2 \times 10^{-5}$ pb at 5 TeV (factor ~ 2), i.e. the falling slope is somewhat shallower than the reference. The SSM/ ψ ratio is roughly constant at ~ 3 across the mass range, close to the reference ordering and spacing.
56. **2103.02708 Fig. 4, claude-opus-4-6:** The reference y-axis is the normalized ratio $((\sigma_B)Z'/(\sigma_B)Z) \times 1928$ pb rather than a raw cross section; agreement is good at high mass (both $\sim 1e-5$ pb for Z'_{SSM} near 5 TeV) but the produced curves sit roughly a factor of a few higher below ~ 1.5 TeV (produced $Z'_{\psi} \approx 0.1$ pb at 1 TeV vs $\sim 2e-2$ in the reference), and the Z'_{ψ} curve stops at ~ 5.0 TeV instead of extending to 5.5 TeV.
57. **2104.05720 Fig. 11, claude-opus-4-6:** Produced curves deviate from SM by at most ~ 2 -3% in any bin for both $m_{LQ} = 1$ TeV and 10 TeV at $\beta_{L^32} = 1$, versus the reference where the 1 TeV curve is shifted by $O(100\%)$ at $|\eta| > 1.2$ and suppressed by a factor ~ 3 in the first bin. The 10 TeV curve and the $\beta_{L^32} = 0.1$ panel agree with the reference at the few-percent level. Normalisation convention also differs (produced uses $1/N dN/d|\eta|$ peaking at ~ 1.5 ; reference plots per-bin normalized fractions peaking at ~ 0.15), which is a bin-width factor, not a shape difference.
58. **2104.05720 Fig. 11, claude-opus-4-6:** In the left panel ($\beta_{L^32} = 1$) the 1 TeV curve peaks at $\eta \approx 1.3$ -1.5 with a normalized height of ≈ 0.118 , slightly broader and $\sim 20\%$ lower than the reference peak of ≈ 0.145 at $\eta \approx 1.1$ -1.3, and it stays somewhat above the reference at large η (≈ 0.04 vs ≈ 0.01 in the last bin). The right panel agrees to within a few percent bin-by-bin. Axis is labelled η rather than η_{μ} .
59. **2104.05720 Fig. 12, claude-opus-4-6:** The produced x-range stops near ~ 50 TeV like the reference but the low-mass end is largely consistent (3 TeV 95% CL ~ 0.035 vs ~ 0.04 , 14 TeV 5-sigma ~ 0.022 vs ~ 0.02 at $m_{LQ} = 1$ TeV); the 14 TeV 95% CL curve sits ~ 2 -3x higher than the reference at low mass (~ 0.017 vs ~ 0.006 at 1 TeV) and shows a spurious bump/dip structure around $m_{LQ} = 1$ -2 TeV that is absent in the smooth reference contours; at the high-mass end the produced 3 TeV curves rise somewhat faster (5-sigma reaching ~ 2 vs ~ 1 near 50 TeV), and the gap between 95% CL and 5-sigma is narrower for the 14 TeV pair than in the reference.
60. **2104.05720 Fig. 12, claude-opus-4-6:** Overall normalisation is close (3 TeV $5\sigma \approx 0.05$ at 1 TeV and ≈ 1 -2 at ~ 50 TeV, matching the reference within tens of percent), but the 14 TeV 95% CL curve sits high at low mass (≈ 0.016 at $m_{LQ} = 1$ TeV versus ≈ 0.006 in the reference), so the exclusion-discovery gap at 14 TeV is compressed relative to the paper; the produced 14 TeV curves also show small non-physical kinks near $m_{LQ} \approx 1.5$ -3 TeV where the reference is smooth.
61. **9909255 Fig. 2, claude-opus-4-6:** Sampled on a coarse 100 GeV grid (200-1200 GeV), so the 600 GeV resonance is sampled rather than resolved: the peak reaches $\sim 3.5e4$ fb versus $\sim 4e4$ fb in the reference, and the interpolated line makes the resonance triangular rather than Breit-Wigner shaped. The scan omits the Z-pole region below 200 GeV shown in the reference and stops at 1200 GeV rather than 1500 GeV, so any second KK resonance near 1100 GeV is not probed. The low-energy end is somewhat high ($\sim 3e3$ fb at 200 GeV versus $\sim 2e3$ fb for the corresponding reference curve), and the 800 GeV dip minimum is shallower/shifted by up to one grid spacing.
62. **9909255 Fig. 2, claude-opus-4-6:** Sampled on a coarse 100 GeV grid from 200-1200 GeV, so the Z pole near 91 GeV and the 1500 GeV endpoint of the reference are not covered, and the 600 GeV resonance peak is captured by a single grid point ($\sim 3 \times 10^4$ fb vs $\sim 4 \times 10^4$ fb for the reference blue curve), understating the true peak height and width. The off-resonance minimum near 400 GeV sits at $\sim 9 \times 10^2$ fb versus $\sim 2 \times 10^3$ fb in the reference, a factor ~ 2 low; the high-energy tail at 1200 GeV ($\sim 2.5 \times 10^4$ fb) agrees with the reference blue curve to within tens of percent.

C Data and code

- docs/paper/results-2026-09-26/: runs.csv (84 attempts), tables.md, longhorizon_light.md, longhorizon_heavy.md, adk_runs.json, figures/ (41), README.md (Chinese narrative). docs/paper/results-2026-09-24/: the first results package (30 figures, documented_runs.json with the 2026-09-16 and 2026-09-24 attempts).
- Harnesses and tools: scripts/bench/run_benchmark.sh, run_benchmark_codex.sh, run_benchmark_adk.sh, run_benchmark_gemini.sh, collect_metrics*.py, judge.sh, judge_all.sh, job_provenance.py, aggregate.py, pack_results.py, longhorizon_table.py, report_tables.py; tests under scripts/tests/.
- Companion documents: claims-audit-2026-09-24.pdf (per-placeholder proposals for the three paper PDFs), handover-2026-09-24.pdf (Chinese handover).